Immanuel Anaborne

571-470-3650 | immanuel.anaborne@gmail.com | linkedin.com/in/anaborne | anaborne.github.io | New York, NY | Available Immediately

Education

University of Pennsylvania Expected December 2026

B.A. Computer Science and English | QuestBridge Match Scholar *Philadelphia, PA*

Experience

Software Engineer, Independent Contractor June 2026 – Present

Mercor, DataAnnotation *Remote*

- Reviewed 2,000+ model-generated Python and JavaScript solutions for logic errors, time complexity and edge cases
- Wrote 500+ multi-turn prompts built to break models on dynamic programming, concurrency and data structures

Data Engineering Research Assistant June 2025 – June 2026

Wharton Accounting Department *Philadelphia, PA*

- Cut manual disclosure-classification effort 50% by building LLM classification pipeline (RAG, OpenAI API) that mapped unstructured corporate filings to trade codes, with event-driven ingestion layer to handle filing updates asynchronously
- Built async ingestion service (asyncio, BeautifulSoup) that pulled and parsed 60,000+ corporate filings, with Pydantic schema validation and pytest suites gating data integrity before load
- Automated entity matching across 500+ inconsistently named firms with vectorized pandas/NumPy ETL workflows

Software Engineer Intern May 2024 – August 2024

GoodRun *New York, NY*

- Replaced paid vendor with in-house React feedback tool, cutting license cost and taking ownership of its data pipeline
- Reduced production deployment failures 15% by hardening CI/CD pipelines with GitHub Actions and rollback rules

Projects

Kalshi NFL/NBA Prop Pricing Study | *Python, pre-registered study* 2026

github.com/anaborne/kalshi-prop-calibration

- Pre-registered a study of Kalshi NFL/NBA prop pricing before pulling any data and published the null: the forecast lost to the exchange mid-quote at −0.0219 Brier skill on the 82.7% of the 87,053-observation training partition quoted two-sided
- Showed the standard mid-quote convention stops being a probability when one side of the book is absent, on 17.3% of the same training partition, and that those rows alone invert the sign of the pooled skill score
- Walked 65,226 settled markets and 1.88M hourly candles unauthenticated; free public data reproduced the exchange's own settlement on 61,978 of 61,984 markets (99.990%), with all six disagreements named

Executor Hot Path in C++ | *C++20, CMake, Catch2, OpenSSL, SQLite* 2026

github.com/anaborne/executor-hotpath-cpp

- Rewrote the trading system's executor process in C++20 against the same wire protocol; the unmodified Python poller drove it across 4,400 frames with every frame accepted and zero telemetry rows dropped
- Timed both executors in one run on one machine: executor-side wake-recv fell 2.4x at p99 and 2.8x at p50 (4.5us to 1.6us at p50), published as an upper bound with the process-topology confound disclosed

Kalshi / Polymarket Trading Infrastructure | *Python, asyncio, uvloop, SQLite, Prefect* 2026

github.com/anaborne/prediction-market-infra

- Architected two-process trading system (poller + pre-warmed executor over Unix domain socket); adopted uvloop over stock asyncio on benchmark evidence, and published the benchmark harness so the comparison runs from a clean clone
- Designed IDF-weighted inverted index that prices two-venue cross in 113s by cutting ∼1B-pair search to ∼63K candidates
- Root-caused position-cap defect that rejected 4,233 valid orders over 23 hours against zero open positions; replaced counter-based estimate with live exchange-position reconciliation at no added API cost

Technical Skills

- **Languages:** Python, C++, TypeScript, JavaScript, SQL
- **Frameworks & Data:** React, Node.js, asyncio, Pydantic, pandas, NumPy, OpenAI API, RAG pipelines, MCP SDK
- **Infrastructure:** Docker, GitHub Actions, Git, pytest, CMake, Catch2

Additional

U.S. National Team Training Partner, Short Track Speed Skating | *US Speedskating* 2019 – 2022

- U.S. Junior Development Team, 2019 and 2020; 16th and 15th overall at the 2019 and 2020 U.S. Championships